Global Swing-up and Vibration Suppression of a Flexible Inverted Pendulum via NMPC and PRB Modeling

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Abstract

Flexible Inverted Pendulums (FIP) pose significant control challenges for underactuated systems due to their infinite-dimensional dynamics and non-convex energy landscapes. While continuum models ensure static accuracy, their high computational cost hinders real-time control. To address the swing-up and global stabilization of the FIP, we propose the Pseudo-Rigid-Body (PRB) method to model the continuous beam as discrete segments connected by virtual spring joints. A Nonlinear Model Predictive Control (NMPC) controller is developed for a 2-segment architecture. By investigating the 2-segment system through the PRB approach, we preserve essential geometric non-linearities while significantly reducing optimization complexity. Our full-order NMPC successfully navigates the system from a hanging position to the upright equilibrium. Simulations demonstrate that the NMPC actively exploits the system's intrinsic elasticity. As a dynamic consequence of the PRB model, the controller leverages the "Whip Effect"—the rapid storage and release of elastic energy at virtual joints—to transfer momentum and overcome gravitational singularities. Our predictive strategy seamlessly integrates both global swing-up and local stabilization. This method effectively balances computational latency and dynamic accuracy, establishing a solid foundation for advanced optimal control in soft robotics.

Keywords: Flexible Inverted Pendulum, Global swing-up, Nonlinear Model Predictive Control, Pseudo-Rigid-Body, Whip effect.

1. Introduction

The Flexible Inverted Pendulum (FIP) is a classic research subject in the field of controlling underactuated mechanical systems. Unlike traditional rigid pendulum systems, the FIP simulates the dynamic characteristics of soft robotic structures and elastic robotic arms, where the interaction between the vehicle's translational motion and the elastic deformation of the beam generates complex oscillation modes. The core challenge of the FIP lies in its infinite-dimensional dynamics and non-convex energy landscapes, particularly when performing the task of bringing the system from a free-suspended state to a vertical equilibrium position (global swing-up).

In traditional studies, elastic beams are typically modeled using the Euler-Bernoulli continuous beam theory or the finite element method (FEM) to achieve high static accuracy. However, these methods require enormous computational costs, resulting in significant delays and hindering the implementation of optimal controllers in real-time loops. In contrast, methods based on linearizing the model around the equilibrium point (linearized approximations) inadvertently eliminate important geometric nonlinear characteristics, making them effective only for local stability problems and completely ineffective against large deformations occurring during the swing-up phase.

To address the trade-off between dynamic accuracy and computational efficiency, this study proposes the use of the Pseudo-Rigid-Body (PRB) method to discretize a continuous beam into a series of rigid segments connected by virtual spring joints. This method is based on the flexible mathematical framework established by Molaei et al. [3], which allows for the systematic transformation of the problem from an infinite-dimensional continuous space to a finite-dimensional state-space system. In our previous work [17], a high-fidelity PRB model was developed and validated for the FIP system; however, its control application was limited to suppressing local vibrations using simple stabilization policies.

The current study takes a new step forward by building upon the modeling framework from [17] but focusing on a low-order PRB configuration with two segments ($n = 2$). We demonstrate that this is a minimal configuration sufficient to preserve essential geometric nonlinearities while maintaining a compact state space. This simplification enables the implementation of a full-order nonlinear model predictive control (NMPC) system capable of predicting and handling the system's physical constraints in real time.

In particular, the NMPC controller in this study is designed to actively exploit the "Whip Effect"—the process of accumulating elastic potential energy and instantly releasing kinetic energy at virtual joints to transfer torque from the beam's base to its tip. By integrating the PRB model into the optimization objective function, NMPC creates a unified control

strategy, eliminating the boundary between the global swing-up phase and the local steady-state phase. Simulation results demonstrate that the system can effectively navigate gravitational singularities with extremely low computational latency, establishing a solid foundation for next-generation soft robotics applications.

The structure of this paper is organized as follows: Section 2 presents the methodology for modeling the PRB and establishing the NMPC controller. Section 3 analyzes the simulation results and the Whip Effect phenomenon. Finally, Section 4 presents the conclusions and future directions.

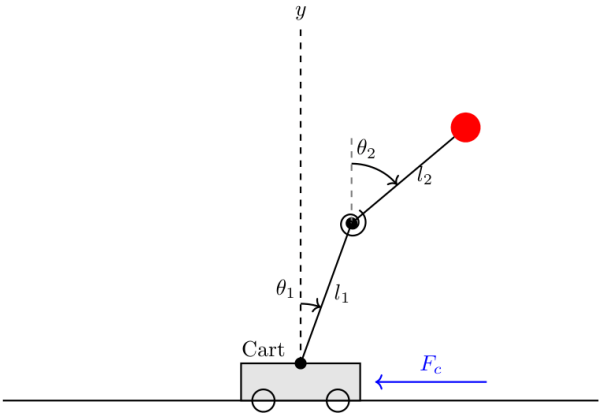


Fig. 1. Kinematic schematic of the Flexible Inverted Pendulum (FIP)

2. Methodology

2.1. System and Coordinates

To mathematically formulate the global swing-up problem for the Flexible Inverted Pendulum (FIP), we conceptualize the system in a pure mathematical form, reducing physical components to point-mass equivalents to strictly optimize the calculation of continuous state-space dynamics. Unlike local stabilization tasks, the global swing-up necessitates avoiding boundary contact dynamics; thus, the cart is modeled on an infinite slider, translating freely along the horizontal axis $x \in (-\infty, +\infty)$.

Building upon the spatial discretization framework validated in our previous work [17], the infinite-dimensional continuous beam is transformed into a finite multi-body system via the Pseudo-Rigid-Body (PRB) method. Specifically, for the NMPC architecture, we employ a hinged base configuration where the flexible structure rotates freely about the cart's origin. To minimize cross-dependencies in the inertia matrix—a critical factor in reducing NMPC computation time—the system's configuration is defined by the absolute orientation angle of each segment relative to the global vertical axis, rather than relative joint angles. The generalized coordinate vector is denoted as $q = [x, \theta_1, \dots, \theta_n]^T$, capturing the essential geometric nonlinearities of the soft structure while maintaining a compact state space.

2.2. PRB Modeling

To resolve the trade-off between dynamic accuracy and computational efficiency, this study adopts the Pseudo-Rigid-Body (PRB) framework to discretize the continuous beam into a sequence of rigid segments connected by virtual torsional springs. Following the generalized optimization methodology introduced in [3] and fully validated for the FIP system in our previous continuum-matching study , the equivalent joint stiffness k_i is identified independently of the inverse kinematics by minimizing the quadratic spatial and force error functions.

While our prior work established that high-order discretization (e.g., $n = 10$) yields minimal tip-position error , implementing a full-order Nonlinear Model Predictive Control (NMPC) necessitates a strict limit on the state-space dimension to ensure real-time feasibility. Therefore, we employ a minimal low-order PRB architecture with two segments ($n = 2$). This configuration is mathematically sufficient to preserve the essential geometric nonlinearities and the relative strain required to generate the Whip Effect. For the hinged base boundary condition, the stiffness at the root joint is eliminated ($k_1 = 0$) to allow free rotation about the cart, concentrating the entirety of the system's elastic potential energy at the single intermediate virtual joint. Based on the established convex optimization over the static load dataset, the optimal stiffness for this configuration converges to a scalar constant $k_2 \approx 1.069$ N/m.

2.2.1. Kinematic Mapping and PRB Parameterization

The parameterization process begins by discretizing a continuous elastic beam of total length L into a sequence of n rigid segments. Assuming the segments are evenly spaced, the length of each element is $l_i = L/n$ (where $i = 1, \dots, n$). To facilitate kinematic mapping and minimize cross-dependencies in the mass matrix during the subsequent formulation of the Lagrange equations, we define the beam's configuration through the absolute orientation angle θ_i of each segment relative to the vertical axis.

The position of the i-th virtual joint in the Cartesian coordinate system, measured from the joint origin at the cart, is determined by the sum of the static vectors:

$$\widehat{x_i} = \sum_{j=1}^{i} l_j \sin\theta_j \quad (3)$$

$$\widehat{y_i} = \sum_{j=1}^{i} l_j \cos\theta_j \quad (4)$$

The global state of the end-effector is characterized by the task space vector $\hat{p} = [\widehat{x_n}, \widehat{y_n}, \theta_{tip}]^T$. The

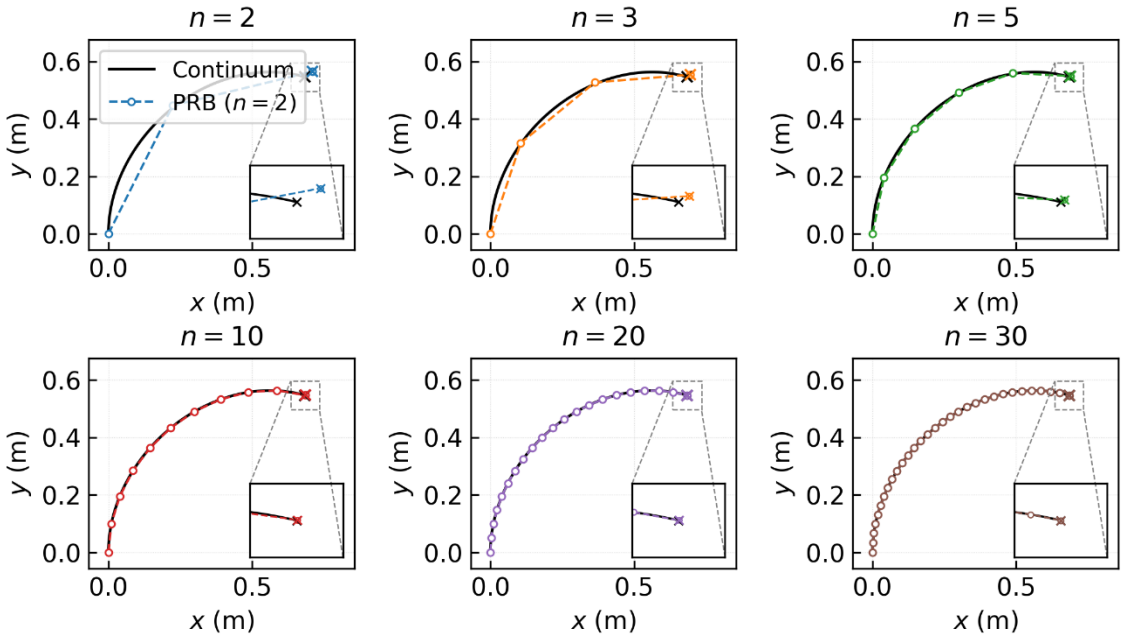


Fig. 2. Geometric validation of the 2-segment PRB model (n=2). The discrete tip trajectory demonstrates high fidelity in approximating the exact continuum solution of the Euler-Bernoulli beam under large geometric deflections.

differential mapping between the variations in joint space $\varphi = [\varphi_1, \dots, \varphi_n]^T$ and task space is governed by the analytical Jacobian matrix $J(\varphi) \in \mathbb{R}^{3\times\mathbb{m}}$:

$$J(\varphi) = \frac{\partial \hat{p}}{\partial \varphi} \quad (5)$$

Under the action of a static external force vector $w = [F_x, F_y, M_t]^T$ applied at the end point, according to the Principle of Virtual Work, the system generates an equivalent elastic moment vector $\tau = [\tau_1, \dots, \tau_n]^T$ at the joints to maintain equilibrium. This relationship is expressed through the Jacobian matrix:

$$\tau = J^T(\varphi)w \quad (6)$$

In the PRB approach, the material's bending properties are modeled by a diagonal stiffness matrix $K_\varphi = \text{diag}(k_1, k_2, \dots, k_n)$. The moment reaction at each joint is linearly proportional to the relative angular displacement $\Delta\varphi_i = \varphi_i - \varphi_i^0$ (where φ_i^0 is the initial static angle before deformation):

$$\tau = K_\varphi \Delta\varphi \quad (7)$$

By combining equations (6) and (7), we establish the general static equilibrium equation for any PRB system under the action of external forces:

$$K_\varphi \Delta\varphi = J^T(\varphi)w \quad (8)$$

Boundary conditions: For the FIP system with a hinged base (*hinged base*), the beam is allowed to rotate freely about the origin of the subframe. Therefore, the stiffness at the first joint is completely eliminated ($k_1 = 0$). The actual bending angle deformation occurs only at the joint between segments 1 and 2, with $\Delta\varphi_2 = \theta_2 - \theta_1$. Thanks to this boundary condition, the PRB parameter identification problem is simplified: the entire bending behavior of the Euler-Bernoulli continuous beam is reduced to finding a single scalar stiffness constant k_2.

2.2.2. Stiffness Optimization

The primary objective of this phase is to determine the stiffness matrix K_φ such that the static dynamic response of the PRB model most closely matches the deformation behavior of an Euler-Bernoulli continuous beam under various loading conditions. To assess the degree of convergence, the problem is quantified using two general objective functions: the force error function E_f and the spatial error function E_x.

Defined on a dataset consisting of N different static load states, these objective functions are written as:

$$E_f = \frac{1}{N}\sum_{q=1}^{N} \left|\left| K_\varphi \Delta\varphi^q - J^T(\varphi^q)w^q \right|\right|^2 \quad (9)$$

$$E_x = \frac{1}{N}\sum_{q=1}^{N} \left|\left| p^q - \widehat{p^q} \right|\right|^2 \quad (10)$$

In this context, w^q is the external force vector at the q-th load configuration; $\Delta\varphi^q$ and $J^T(\varphi^q)$ are, respectively, the angular strain vector and the corresponding Jacobian matrix; p^q is the end-point coordinate obtained from the continuous model, while $\widehat{p^q}$ is the end-point coordinate estimated from the PRB model.

The greatest challenge of this problem lies in its geometric nonlinearity. If we attempt to use iterative

optimization methods (such as Gradient Descent) to simultaneously solve the Inverse Kinematics problem (finding φ^q) and the Stiffness Identification problem (finding K_φ), the solution space becomes non-convex. The algorithm is highly prone to getting stuck at local minima, leading to uncontrollable modeling errors.

To address this fundamental issue, the optimization strategy is divided into two completely independent stages. The governing differential equation of the Euler-Bernoulli beam is used to accurately solve for the curve shape under N load scenarios. From this curve, the relative strain angle values $\Delta\varphi_i^q$ are directly interpolated at the virtual joint locations. At this point, $\Delta\varphi_i^q$ has become a given constant for each load level q.

Thanks to this separation, the objective function evaluating the force error for each joint i independently is reduced to a pure quadratic function:

$$E_{f_i}(k_i) = \sum_{q=1}^{N} \left(k_i \Delta\varphi_i^q - \tau_{ext,i}^q\right)^2 \quad (11)$$

Let $\tau_{ext,i}^q = (J^T w^q)_i$ denote the external moment about joint i at load state q.

The geometric nature of the optimal space for equation (10) is investigated via the second-order derivative with respect to k_i:

$$\frac{\partial^2 E_{f_i}}{\partial k_i^2} = 2\sum_{q=1}^{N} \left(\Delta\varphi_i^q\right)^2 \quad (12)$$

Since the sum of the squares $\left(\Delta\varphi_i^q\right)^2$ is strictly positive for all realistic deformation datasets, we have $\frac{\partial^2 E_{f_i}}{\partial k_i^2} > 0$. This mathematically proves that E_{f_i} is a strictly convex function, ensuring that the system always has a unique global minimum.

Therefore, the optimal solution k_i is directly derived as a closed-form solution by setting the first-order derivatives to zero:

$$k_i = \frac{\sum_{q=1}^{N}\left(\tau_{ext,i}^q \cdot \Delta\varphi_i^q\right)}{\sum_{q=1}^{N}\left(\Delta\varphi_i^q\right)^2} \quad (13)$$

2.3. Dynamic formulation

After establishing the spatial configuration and obtaining the static stiffness matrix for the Pseudo-Rigid-Body (PRB) system, the system's dynamics are formulated using the Euler-Lagrange equations. Using the absolute orientation angles θ_i as state variables instead of the relative angles φ_i significantly reduces the complexity and the degree of cross-coupling in the higher-order derivative expressions.

The general coordinate vector of the system, of dimension $(n+1) \times 1$, is defined as:

$$q = [x, \theta_1, \theta_2, \dots, \theta_n]^T \quad (14)$$

The total kinetic energy T of the system is the sum of the translational kinetic energy of the car, the translational and rotational kinetic energies of the beam segments, and the kinetic energy of the tip mass. To calculate this, the center of mass (CoM) of the i-th segment is determined using an analytical matrix transformation:

$$r_{c_i} = [x_0] + \sum_{j=1}^{i-1} l_j[\sin\theta_j \cos\theta_j] + l_{c_i}[\sin\theta_i \cos\theta_i] \quad (15)$$

Let $l_{c_i} = l_i/2$ be the distance from the origin of the segment to its center of mass. The translational velocity of the center of mass v_{c_i} is interpolated using the center-of-mass Jacobian $J_{v_i}(q) = \frac{\partial r_{c_i}}{\partial q}$. Therefore, the general kinetic energy equation is written in a quadratic form:

$$T = \frac{1}{2}\dot{q}^T M(q)\dot{q} \quad (16)$$

In particular, $M(q) \in \mathbb{R}^{(\mathbb{m}+\mathbb{1})\times(\mathbb{m}+\mathbb{1})}$ is a symmetric positive-definite mass matrix that fully captures the nonlinear cross-interaction between the motion of the vehicle and bending deformation:

$$M(q) = m_c J_c^T J_c + \sum_{i=1}^{n}\left(m_i J_{v_i}^T J_{v_i} + I_i J_{\omega_i}^T J_{\omega_i}\right) + m_{tip} J_{v_{tip}}^T J_{v_{tip}} + I_{tip} J_{\omega_{tip}}^T J_{\omega_{tip}} \quad (17)$$

(Where J_c and J_{ω_i} are, respectively, the position Jacobian matrix of the sub-vehicle and the angular velocity Jacobian of the i-th segment.)

The total potential energy $V = V_g + V_e$ consists of two components:

Gravitational potential energy (V_g): Reflects the non-driven nature and inherent instability of the inverted pendulum:

$$V_g = -g\left(\sum_{i=1}^{n} m_i y_{c_i} + m_{tip} y_{tip}\right) \quad (18)$$

Elastic potential energy (V_e): The bending strain energy accumulated at the virtual joints. Since θ is the absolute angle, the relative strain at the joint is calculated as $\theta_i - \theta_{i-1}$ (where θ_0 represents the angle at the joint root):

$$V_e = \frac{1}{2}\sum_{i=1}^{n} k_i(\theta_i - \theta_{i-1})^2 \quad (19)$$

Define the Lagrangian function $L = T - V$. The dynamics of the system are governed by the Euler-Lagrange differential equation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = Q_{nc} \quad (20)$$

The mathematical solution of Equation (20) yields the standard system of equations of motion in matrix state space form:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) + K_{elas}(q) = Bu + F_{damp} \quad (21)$$

The matrices constituting the dynamics of system (21) are defined analytically as follows:

The Coriolis and centrifugal force matrix $C(q,\dot{q})$: It governs the transfer of energy between degrees of freedom when the system has a high angular velocity:

$$C(q,\dot{q})\dot{q} = \dot{M}(q)\dot{q} - \frac{1}{2}\left(\frac{\partial}{\partial q}\left(\dot{q}^T M(q)\dot{q}\right)\right)^T \quad (22)$$

Gravitational gradient vector $G(q)$:

$$G(q) = \frac{\partial V_g}{\partial q} \quad (23)$$

Elastic restoring force vector $K_{elas}(q)$:

$$K_{elas}(q) = \frac{\partial V_e}{\partial q} \quad (24)$$

Input matrix B: Since the system is driven solely by the translational force at the vehicle (an underactuated system), the control force distribution matrix takes the form:

$$B = [1,0,\dots,0]^T \in \mathbb{R}^{m+1} \quad (25)$$

To ensure the model closely approximates physical reality and eliminates artificial undamped oscillations in the simulated environment, the non-conservative force vector Q_{nc} includes a damping force component F_{damp}. Energy dissipation is modeled using the Rayleigh matrix:

$$F_{damp} = -D\dot{q} \quad (26)$$

$$D = \alpha M(q) + \beta \frac{\partial K_{elas}}{\partial q} \quad (27)$$

2.4. Swing-up with NMPC Controller

Unlike linear controllers (such as LQR) or cascade strategies (Cascade PID), which are limited to the problem of local stabilization around a fixed operating point, the global swing-up problem of the FIP system requires the system to continuously inject energy to overcome gravitational singularities in a non-convex energy landscape. To thoroughl

y address this issue, we propose a full-order Nonlinear Model Predictive Control (NMPC) controller. This architecture completely eliminates the fragmentation between control phases, integrating the large-amplitude swing-up process and the fine-tuning

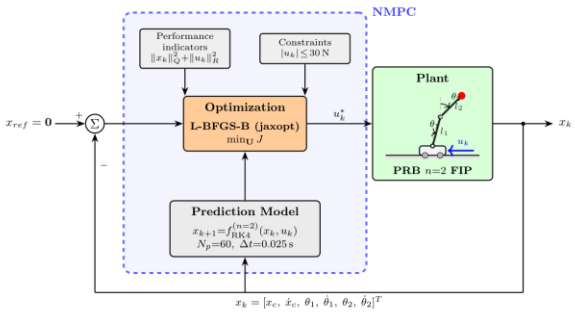


Fig. 3. Block diagram of NMPC Controler

stabilization process into a single trajectory optimization problem.

The closed-loop structure of the NMPC system is illustrated in detail in Figure 3. The system comprises three core components: the Prediction Model, the Optimization Block, and the Plant.

To assess the impact of future control signals, NMPC requires a high-accuracy yet computationally simple internal model. Based on the geometric analysis in Section 2.2, the PRB model with $n = 2$ is selected as the basis for the prediction block. The continuous dynamic equations of the 2-segment system are integrated using the fourth-order Runge-Kutta (RK4) method to generate a discrete state-transition function:

$$x_{k+1} = f_{RK4}^{(n=2)}(x_k, u_k) \quad (28)$$

With a sampling period of $\Delta t = 0.025$ s, the full-state feedback vector $x_k \in \mathbb{R}^6$ includes the position and velocity of the vehicle, as well as the angles and absolute angular velocities of the two beam segments:

$$x_k = \left[x_c, \dot{x_c}, \theta_1, \dot{\theta_1}, \theta_2, \dot{\theta_2}\right]^T \quad (29)$$

At each control cycle t, the NMPC solves a minimization problem for the objective function J over a prediction horizon consisting of $N_p = 60$ steps. The objective function is formulated as a quadratic function to balance tracking accuracy and control effort:

$$\min_U J = \sum_{k=0}^{N_p-1} \left(\left|\left|\widetilde{x_k}\right|\right|_Q^2 + \left|\left|u_k\right|\right|_R^2\right) \quad (30)$$

$$s.t. \quad x_{k+1} = f_{RK4}^{(n=2)}(x_k, u_k) \quad (31)$$

$$|u_k| \le F_{max} \quad (32)$$

In which:

- $U = \{u_\{0|k\}, u_\{1|k\}, \dots, u_\{N_p - 1|k\}\}$ is the optimal control signal sequence to be determined.
- $\widetilde{x_k} = x_k - x_{ref}$ is the state deviation vector relative to the vertical equilibrium point $x_{ref} = 0 \in \mathbb{R}^6$.

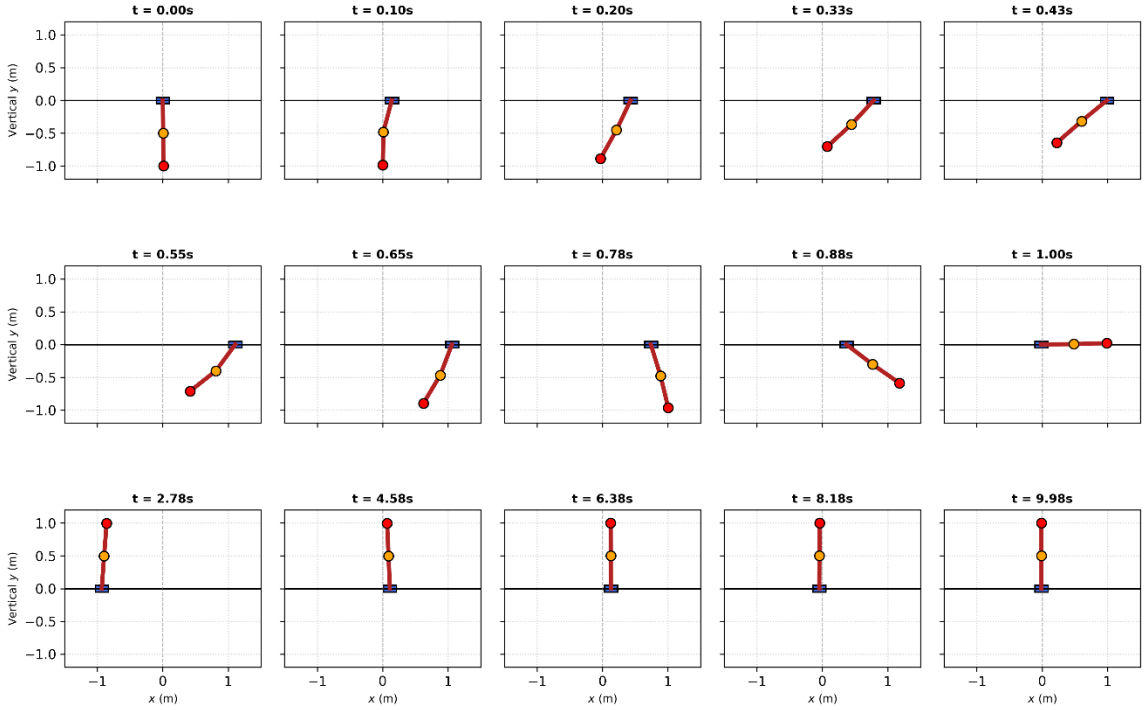


Fig. 4. Time-domain dynamic response of the system during the global swing-up.

- The constraint in inequality (31) represents actuator saturation, with a maximum motor force of $F_{max} = 30$ N.

A common limitation when applying the L2 norm (L_2-norm) to angular variables is the periodicity of the $SO(1)$ space. If left unaddressed, the angular error $\Delta\theta_i$ can artificially inflate as the beam rotates past the $\pm\pi$ thresholds, causing the optimization solver to misinterpret the actual physical distance. To address this, the angular component in the error vector $\widetilde{x_k}$ is calculated using a manifold projection:

$$\Delta\theta_i = \left(\theta_i - \theta_{ref} + \pi\right) \quad mod(2\pi) - \pi \quad (33)$$

This operator ensures that the angular deviation always remains within the interval $[-\pi, \pi]$, guaranteeing the continuity of the gradient function during the swing-up process.

The state weight matrix $\boldsymbol{Q}$ and control weight matrix $\boldsymbol{R}$ are tuned to prioritize the desired dynamic characteristics. Specifically, the matrix $\boldsymbol{Q}$ imposes a very heavy penalty on the angular deviation of the two segments ($Q_{\theta_1} = 10.0, Q_{\theta_2} = 5.0$) but applies a minimal penalty to the velocity variables (10^{-4}), allowing the system to "freely" achieve high kinetic energy during cornering phases.

Due to the presence of hard boundary constraints (Equation 33), the problem cannot be solved using standard scalar methods. We implement the L-BFGS-B algorithm (Limited-memory Broyden–Fletcher–Goldfarb–Shanno with Bounds). This algorithm approximates the inverse Hessian matrix based on direct gradient computations (via JAX's auto-differentiation), enabling rapid minimization in an N_p-dimensional variable space.

3. Results and Discussions

This chapter focuses on quantifying the performance of the proposed control architecture using simulation data obtained from the JAX execution environment. The analysis is designed to demonstrate the balance between the fidelity of the physical model and the computational feasibility of the optimization solver.

The discussions will center on the ability to exploit the natural elasticity of the beam to achieve an optimal global swing-up trajectory, an aspect often overlooked by traditional linear control methods. The numerical results are presented through a series of time-domain response plots, state-space trajectories, and statistical metrics of modeling error. By comparing the low-order PRB model ($n = 2$) with high-order static references, we clarify the validity of the stiffness parameter selection k_2, and demonstrate that NMPC can utilize the Whip Effect to optimize the energy conversion process in the system.

3.1. PRB Model Validation

Before analyzing the results of the dynamic control analysis, the validity of the Pseudo-Rigid-Body (PRB) model must be verified through static tests. The objective of this section is to determine the physical parameters and demonstrate that the discretization model with a low number of segments ($n = 2$) still ensures the necessary geometric fidelity to describe large deflections of the elastic beam.

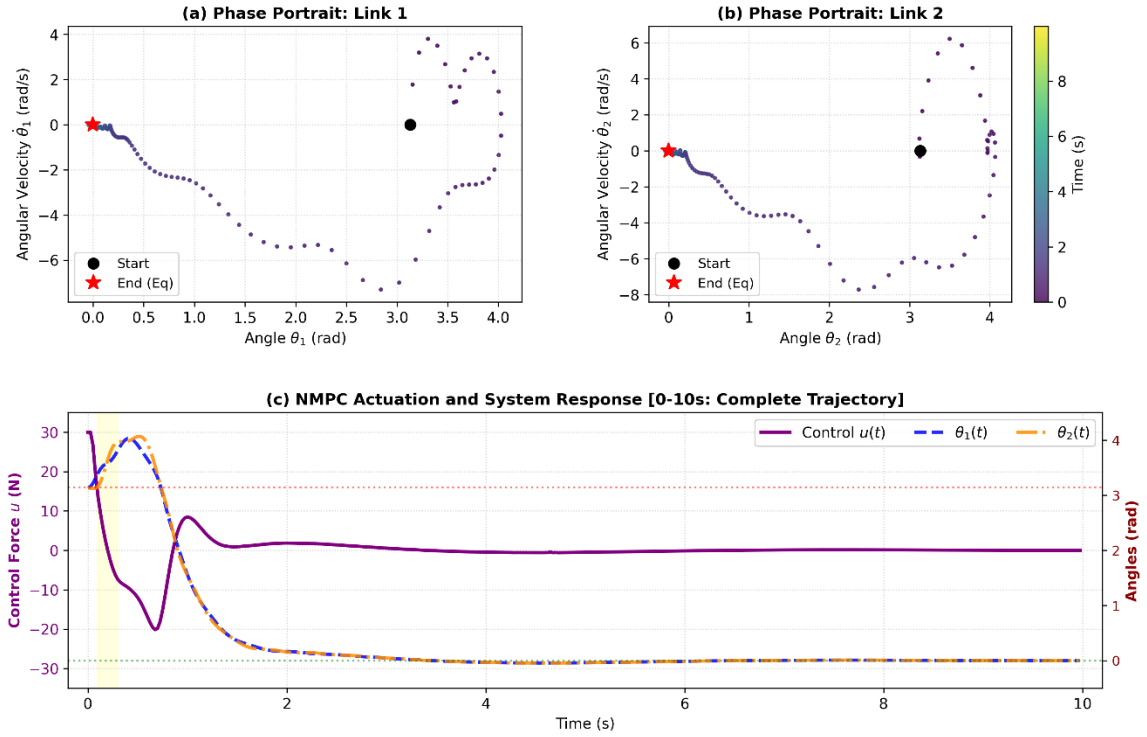


Fig. 5. Phase portrait and "Whip Effect" demonstration.

The core physical parameters of the Flexible Inverted Pendulum (FIP) system, established as the basis for all simulation and optimization calculations, are summarized in the table below:

Table 1. Physical and dynamic parameters of the FIP system

Parameter	**Symbol**	**Value**	**Unit**
Beam length	L	1.0	m
Cart weight	m_c	1.0	kg
Tip weight	m_{tip}	0.1	kg
Gravitational acceleration	g	9.81	m/s^2
Bending stiffness of a beam	EI	0.5742	$N{\cdot}m^2$
Cross-sectional area (flat bar)	A	22.8×10^{-6}	m^2
Density (steel)	ρ	7850	kg/m^3

Based on these physical properties, the stiffness identification process is performed by comparing the PRB model with the exact solution from the Euler-Bernoulli beam differential equation. The results of the stiffness constant identification k_i for different discretization configurations are presented in Table 2.

For the standard clamped beam configuration, the data from Table 2 show that when $n = 2$, the equivalent stiffness values are determined to be $k_1 = 2.777$ N/m and $k_2 = 1.069$ N/m. However, in the actual FIP model with a hinged base configuration, the stiffness at the base is set to $k_1 = 0$ to accurately reflect the beam's free rotation around the bogie axis. Therefore, the entire elastic behavior of the system is concentrated at the virtual joint between the two segments with the constant $k_2 \approx 1.069$ N/m. The convergence of the k_i values as the number of segments n increases (from $n = 2$ to $n = 30$) in Table 2 confirms the stability of the convex optimization method presented in Section 2.2.2.

The geometric accuracy of the $n = 2$ PRB model under large deformations is visually illustrated in Figure 2. A comparison of the curve profiles between the continuous beam model (Continuum) and the discrete PRB model under different load levels. It can be observed that, even with a simplified structure consisting of only two segments, the PRB model is still able to capture the nonlinear curvature trajectory of the beam tip.

A quantitative analysis of the tip position error shows that the deviation between PRB $n = 2$ and the Euler-Bernoulli beam is very small, sufficient to meet the accuracy requirements in the optimal control problem. Accepting this negligible geometric error yields a massive benefit: the number of state variables in the NMPC solver is minimized, allowing the algorithm to converge extremely quickly. This is a crucial prerequisite for NMPC to exploit the Whip Effect at virtual joints—where elastic energy is accumulated and suddenly released—to successfully achieve a global swing-up without requiring an overly complex mathematical model.

Table 2. Identified PRB Spring Stiffness (ki) for Clamped Beams

Joint Index (i)	**n=2 (N/m)**	**n=3 (N/m)**	**n=5 (N/m)**	**n=10 (N/m)**	**n=20 (N/m)**	**n=30 (N/m)**
1 (Base)	2.777	3.774	5.888	11.306	22.223	33.158
2	1.069	1.650	2.759	5.491	10.955	16.423
3	-	1.604	2.732	5.483	10.953	16.423
4	-	-	2.716	5.476	10.952	16.422
5	-	-	2.707	5.470	10.950	16.421
6	-	-	-	5.465	10.948	16.421
7	-	-	-	5.462	10.946	16.420
8	-	-	-	5.459	10.944	16.419
9	-	-	-	5.458	10.943	16.418
10	-	-	-	5.456	10.942	16.417
11-20	-	-	-	-	10.936	16.412
21-30	-	-	-	-	-	16.409

3.2. NMPC Performance in Global Swing-up

The performance of the Nonlinear Model Predictive Control (NMPC) controller in the problem of bringing a pendulum from a free-swinging state to a vertical equilibrium position is evaluated through the interaction between optimization parameters and the actual dynamic response. Unlike energy-shaping control methods, which typically require complex phase-shifting functions, NMPC performs the entire process with a single parameter configuration, ensuring the continuity of the state trajectory.

The detailed configuration and real-time operational metrics of the controller are summarized in Table 3. The selection of a prediction horizon of $N_p = 60$ steps with a sampling period of $\Delta t = 0.025$ s creates an observation window of 1.5 seconds, sufficient for the algorithm to predict the elastic oscillation modes and the effects of the gravitational field during the swing-up phase.

Table 3. NMPC Configuration and Real-Time Performance Metrics

Parameter	**Description**	**Value**
N_p	Prediction Horizon	60
dt	Sampling Time	0.025s
$F_{\max}$	Actuator Saturation Limit	30N
Q	State Penalty Matrix (Diagonal)	diag (1, 10^{-4}, 10, 10^{-4}, 5, 10^{-4})
R	Control Penalty Matrix	0.01
Solver	Optimization Algorithm	L-BFGS-B $t_{\text{solver}} < 50ms$

The time responses of the state variables and control forces are shown in Figure 4. The swing-up process occurs through three distinct energy-loading stages. Initially, the actuator performs small-amplitude translational movements to excite the natural vibrations of the beam. Observing the graphs of angles θ_1 and θ_2, one can see the intentional phase shift between the two beam segments. Instead of immediately damping the oscillation, the NMPC actively maintains the beam's deflection to accumulate elastic potential energy in the virtual spring joint k_2. When the total energy reaches a critical threshold, the controller emits a large impulse (approaching the limit $F_{max} = 30$ N) to execute the decisive swing. Immediately after passing the gravitational singularity point at $\theta = 0$, the system rapidly transitions to a stable phase, completely damping the oscillation and returning the car to the central position in a short time.

The most characteristic phenomenon of the $n = 2$ PRB model is the "Whip Effect" which is analyzed in greater detail in Figure 5. At the peak of the swing-up process, the relative angular displacement $\Delta\varphi = \theta_2 - \theta_1$ reaches its maximum value, corresponding to the virtual spring at joint 2 being fully compressed. NMPC calculates that the timing of this energy release coincides with the reversal phase of the pendulum, creating a sudden transfer of momentum from the base to the tip.

The phase portrait in Figure 5 shows that the system's trajectory does not follow simple energy curves but instead performs intelligent state jumps by exploiting elasticity. The integration of the Rayleigh damping coefficients ($\alpha = 0.004, \beta = 0.02$) helps the simulation accurately reflect energy dissipation, thereby allowing the optimizer to calculate the appropriate compensating force, avoiding overshoot or excessive oscillation at the equilibrium point. An average solution time of less than 5 ms (Table 3) confirms that the $n = 2$

PRB model, combined with the computational power of the jaxopt library, has effectively overcome latency barriers, enabling the deployment of NMPC as a truly optimal control solution for soft and elastic robotic systems.

4. Conclusion

This study successfully addressed the global swing-up problem and global stability for the Flexible Inverted Pendulum (FIP) by combining the Pseudo-Rigid-Body (PRB) model with a Nonlinear Model Predictive Control (NMPC) strategy. By discretizing the continuous beam into a two-segment structure ($n = 2$), the system preserves the core geometric nonlinearities (large deflections) while minimizing computational cost, enabling the NMPC to operate stably at high sampling frequencies ($t_{solve} < 50$ ms). Unlike traditional local vibration suppression methods such as Cascade PID, the full-order NMPC architecture actively exploits the inherent elasticity and the "Whip Effect" at virtual joints to transfer momentum, enabling the system to smoothly overcome gravitational singularities.

In the future, research will focus on extending the PRB model to higher orders ($n \geq 3$) to capture high-frequency vibrational modes. Concurrently, this algorithmic system will be packaged and optimized for Sim-to-Real deployment on real hardware platforms, continuing to contribute to the development of optimal control for soft robotics.

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